

The goal of this document is to derive the derivative of a quadratic matrix equation when some values of x are not variables and are instead set to 0. We want to see if there is a simple way to move between the regimes where we take the derivative of x with respect to all or some of its components.

$$y = x^T A x$$

At the end, we will extend this to the case where X is a matrix as in the Kalman filter tutorial $y = \text{Tr}[X^T A X]$

NOTE that for the Kalman filter tutorial x corresponds to the system weights (one if which is called A sorry) and y and A correspond to sufficient statistics.

Strategy for derivation of $\frac{\partial y}{\partial x}$:

1. expand $x^T A x$
2. Take the derivative of this equation with respect to each x_i
3. Consolidate into matrix form

Step 1

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}^T \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}^T \begin{bmatrix} A_{11}x_1 + A_{12}x_2 + A_{13}x_3 \\ A_{21}x_1 + A_{22}x_2 + A_{23}x_3 \\ A_{31}x_1 + A_{32}x_2 + A_{33}x_3 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}^T \begin{bmatrix} A_{11}x_1 + A_{12}x_2 + A_{13}x_3 \\ A_{21}x_1 + A_{22}x_2 + A_{23}x_3 \\ A_{31}x_1 + A_{32}x_2 + A_{33}x_3 \end{bmatrix} = A_{11}x_1^2 + A_{12}x_1x_2 + A_{13}x_1x_3 + A_{21}x_1x_2 + A_{22}x_2^2 + A_{23}x_2x_3 + A_{31}x_1x_3 + A_{32}x_2x_3 + A_{33}x_3^2$$

Step 2

$$\begin{bmatrix} \frac{\partial y}{\partial x_1} \\ \frac{\partial y}{\partial x_2} \\ \frac{\partial y}{\partial x_3} \end{bmatrix} = \begin{bmatrix} 2A_{11}x_1 + A_{12}x_2 + A_{13}x_3 + A_{21}x_2 + A_{31}x_3 \\ A_{12}x_1 + A_{21}x_1 + 2A_{22}x_2 + A_{23}x_3 + A_{32}x_3 \\ A_{13}x_1 + A_{23}x_2 + A_{31}x_1 + A_{32}x_2 + 2A_{33}x_3 \end{bmatrix}$$

$$\begin{bmatrix} \frac{\partial y}{\partial x_1} \\ \frac{\partial y}{\partial x_2} \\ \frac{\partial y}{\partial x_3} \end{bmatrix} = \begin{bmatrix} A_{11}x_1 + A_{21}x_2 + A_{31}x_3 \\ A_{12}x_1 + A_{22}x_2 + A_{32}x_3 \\ A_{13}x_1 + A_{23}x_2 + A_{33}x_3 \end{bmatrix} + \begin{bmatrix} A_{11}x_1 + A_{12}x_2 + A_{13}x_3 \\ A_{21}x_1 + A_{22}x_2 + A_{23}x_3 \\ A_{31}x_1 + A_{32}x_2 + A_{33}x_3 \end{bmatrix}$$

Step 3

Note that the above is equal to

$$\frac{\partial y}{\partial x} = Ax + A^T x = (A + A^T)x$$

Now calculate the Derivation of $\frac{\partial y}{\partial x}$ when x_1 is 0 :

$$y = x^T A x$$

Step 1

$$\begin{bmatrix} 0 \\ x_2 \\ x_3 \end{bmatrix}^T \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} \begin{bmatrix} 0 \\ x_2 \\ x_3 \end{bmatrix} = A_{22}x_2^2 + A_{23}x_2x_3 + A_{32}x_2x_3 + A_{33}x_3^2$$

Step 2

note, you do not take the derivative with respect to x_1 because it is not a variable

$$\begin{bmatrix} \frac{\partial y}{\partial x_2} \\ \frac{\partial y}{\partial x_3} \end{bmatrix} = \begin{bmatrix} 2A_{22}x_2 + A_{23}x_3 + A_{32}x_3 \\ A_{23}x_2 + A_{32}x_2 + 2A_{33}x_3 \end{bmatrix}$$

$$\begin{bmatrix} \frac{\partial y}{\partial x_2} \\ \frac{\partial y}{\partial x_3} \end{bmatrix} = \begin{bmatrix} A_{22}x_2 + A_{32}x_3 \\ A_{23}x_2 + A_{33}x_3 \end{bmatrix} + \begin{bmatrix} A_{22}x_2 + A_{23}x_3 \\ A_{32}x_2 + A_{33}x_3 \end{bmatrix}$$

Step 3

$$\frac{\partial y}{\partial \tilde{x}} = \tilde{A}\tilde{x} + \tilde{A}^T\tilde{x} = (\tilde{A} + \tilde{A}^T)\tilde{x}$$

where

$$\tilde{A} = \begin{bmatrix} A_{22} & A_{23} \\ A_{32} & A_{33} \end{bmatrix}, \tilde{x} = \begin{bmatrix} x_2 \\ x_3 \end{bmatrix}$$

Conclusion :

When $y = x^\top Ax$, $\frac{\partial y}{\partial x} = (A + A^\top)x$

Consider the case where some entry i in x is not a variable and is set to 0

$\frac{\partial y}{\partial \tilde{x}} = (\tilde{A} + \tilde{A}^\top)\tilde{x}$ where

$\tilde{A}$ is equal to A with column and row i removed

$\tilde{x}$ is equal to x with element i removed

Extra notes :

A similar strategy of removing rows and columns works for calculating $\frac{\partial y}{\partial x}$ when $y = Ax$

This also works if x is a matrix and $y = \text{Tr}[X^\top AX]$

$$y = \text{Tr} \left[\begin{bmatrix} X_{[:,1]} & X_{[:,2]} \end{bmatrix}^\top A \begin{bmatrix} X_{[:,1]} & X_{[:,2]} \end{bmatrix} \right]$$

$$y = \text{Tr} \left[\begin{bmatrix} X_{[:,1]}^\top A \\ X_{[:,2]}^\top A \end{bmatrix} \begin{bmatrix} X_{[:,1]} & X_{[:,2]} \end{bmatrix} \right]$$

$$y = \text{Tr} \left[\begin{bmatrix} X_{[:,1]}^\top AX_{[:,1]} & X_{[:,1]}^\top AX_{[:,2]} \\ X_{[:,2]}^\top AX_{[:,1]} & X_{[:,2]}^\top AX_{[:,2]} \end{bmatrix} \right]$$

$$y = X_{[:,1]}^\top AX_{[:,1]} + X_{[:,2]}^\top AX_{[:,2]}$$

$$\frac{\partial y}{\partial X_{[:,i]}} = (A + A^\top)X_{[:,i]}$$

Since each column of $\frac{\partial y}{\partial \tilde{X}}$ is calculated independently, you can still use our derived formula for setting values of x to 0

$$\frac{\partial y}{\partial \tilde{X}_{[:,i]}} = (\tilde{A} + \tilde{A}^\top)\tilde{X}_{[:,i]}$$